

The algebraic small object argument as a saturating operation

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joint work in progress with Christian Sattler

Weak factorization systems

- ⊗ On a category $\mathcal{C}$, pair $(\mathcal{L}, \mathcal{R})$ of $\mathcal{L}, \mathcal{R} \subseteq \text{Ob } \mathcal{C}^{\rightarrow}$



- ⊗ Often generated by a set $\mathcal{S} \subseteq \text{Ob } \mathcal{C}^{\rightarrow}$ of left maps

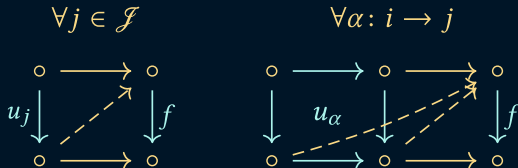
$$\mathcal{R} = \{\text{maps right lifting against all } f \in \mathcal{S}\}$$

- ⊗ (complemented mono, split epi) in **Set** generated by $\{0 \rightarrowtail 1\}$

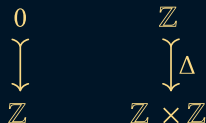
- ⊗ (triv cofib, Kan fib) on $\text{PSh}(\Delta)$ generated by $\{\Lambda_k^n \rightarrowtail \Delta^n \mid k \leq n \in \mathbb{N}\}$

Generation by a category

⊗ f right lifts against $u : \mathcal{J} \rightarrow \mathcal{E}^{\rightarrow}$ when



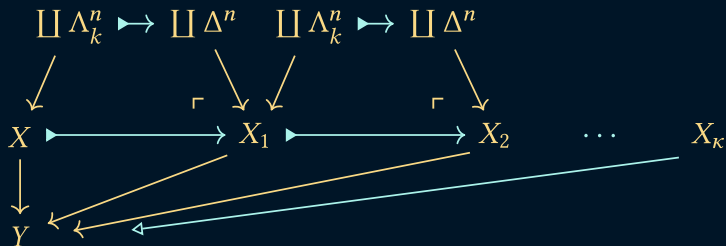
⊗ (complemented mono, split epi) in **AbGrp** generated by full subcategory of $\mathcal{E}^{\rightarrow}$ containing



⊗ “uniform fibrations” in, e.g., cubical sets (used in models of HoTT)

Quillen's small object argument

- ⊗ Build a WFS from a set $\mathcal{S}$ of generators (assuming ...)



- ⊗ The left factor is a transfinite composite of cobase changes of coproducts of generating maps
- ⊗ Any left map is a retract (in $\mathcal{E}^{\rightarrow}$) of one of these

Saturation

Def. $\mathcal{A} \subseteq \text{Ob } \mathcal{E}^{\rightarrow}$ is saturated when it is closed under coproducts, cobase change, transfinite composition, and retracts.

- ⊗ When $(\mathcal{L}, \mathcal{R})$ is generated from $\mathcal{S}$ by the SOA, $\mathcal{L}$ is the least saturated class containing $\mathcal{S}$.

Saturation for categories?

- ⊗ Garner's algebraic small object argument (2008)
generates a wfs given $u: \mathcal{J} \rightarrow \mathcal{E}^{\rightarrow}$
- ⊗ Builds left factors as transfinite composites

$$X \longrightarrow X_1 \longrightarrow X_2 \longrightarrow \cdots \longrightarrow X_{\kappa}$$

... but step maps may not be left maps!

- ⊗ 💡: see

$$\begin{array}{ccccccc} X & \xlongequal{\quad} & X & \xlongequal{\quad} & X & \xlongequal{\quad} & \cdots \xlongequal{\quad} X \\ \Downarrow & & \Downarrow & & \Downarrow & & \Downarrow \\ X & \longrightarrow & X_1 & \longrightarrow & X_2 & \longrightarrow & \cdots \longrightarrow X_{\kappa} \end{array}$$

as a colimit in the category **LeftMap** of left-structured maps

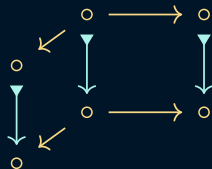
Result

⊗ Left factors are built by colimits of diagrams in **LeftMap**:

(a) sequential colimits



(b) pushouts



(c) colimits of diagrams factoring through $\mathcal{J}$

$$\mathcal{J} \xrightarrow{d} \mathcal{J} \xrightarrow{u} \mathbf{LeftMap}$$

⊗ and “vertical” composition $\circ \blacktriangleright \longrightarrow \circ \blacktriangleright \longrightarrow \circ$

Wrap-up

- ⊗ Can parlay idea into universal property of the double category of left-structured maps
- ⊗ Under conditions on $u: \mathcal{J} \rightarrow \mathcal{E}^{\rightarrow}$, form of colimits can be further constrained, *cf.* Athorne's coalgebraic cell complexes (2014)
- ⊗ Saturation principle for wfs generated by a set is a special case

POST-TALK UPDATE:

This slide was too optimistic. We:

- prove a partial universal property, giving existence but not uniqueness
- do not know if we can recover the standard principle for a WFS generated by a set as a special case.

For more details, see the preprint: <https://arxiv.org/abs/2506.02759>

Thank you!

References

- [1] Thomas Athorne. “Coalgebraic Cell Complexes”. PhD thesis. University of Sheffield, 2014. URL: etheses.whiterose.ac.uk/6285.
- [2] John Bourke and Richard Garner. “Algebraic weak factorisation systems I: Accessible AWFS”. In: *Journal of Pure and Applied Algebra* 220.1 (2016), pp. 108–147. DOI: [10.1016/j.jpaa.2015.06.002](https://doi.org/10.1016/j.jpaa.2015.06.002).
- [3] Richard Garner. “Understanding the small object argument”. In: *Applied Categorical Structures* 17 (2009), pp. 247–285. DOI: [10.1007/s10485-008-9137-4](https://doi.org/10.1007/s10485-008-9137-4).
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- [5] Jiří Rosický. “Accessible Model Categories”. In: *Applied Categorical Structures* 25 (2017), pp. 187–196. DOI: [10.1007/s10485-015-9419-6](https://doi.org/10.1007/s10485-015-9419-6).